Cross-Lingual Transfer Learning in NLP: Enhancing English Language Learning for Non-Native Speakers

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***Abstract*--In the evolving landscape of language education, this paper delves into the intersection of Natural Language Processing (NLP) and education technology to address the unique challenges faced by non-native speakers, particularly children, in acquiring English proficiency. Leveraging the potential of Cross-Lingual Transfer Learning, the proposed methodology, implemented in Python, aims to enhance language learning outcomes through the innovative use of the NNCES (NNCES) corpus. The NNCES corpus, featuring 50 Telugu-speaking children aged 8 to 12 engaged in English language learning, serves as a rich dataset for exploring cross-lingual transfer learning strategies. The paper introduces Multilingual Transfer Learning with Domain Adaptation (MTL-DA) that fine-tunes pre-trained multilingual models on the NNCES corpus. This strategic adaptation enables models to discern linguistic nuances, phonetic variations, and semantic context inherent in NNCES. The methodology involves a comprehensive pipeline, encompassing dataset collection, preprocessing, feature extraction, and model training. Min-Max Normalization is applied to acoustic features, and Mel-frequency cepstral coefficients (MFCCs) and word embeddings from pre-trained models are integrated into a holistic feature vector. The fine-tuned models exhibit superior performance in English language learning tasks, showcasing an accuracy of 99.6%. Comparative analysis with existing methods reveals a significant improvement, with the proposed method surpassing others by 3.6%.**

Keywords--Natural Language Processing, Cross-Lingual Transfer Learning, Multilingual Transfer Learning; Domain Adaptation, Non-Native Speakers, Mel-Frequency Cepstral Coefficients

I. INTRODUCTION

The landscape of language education is globally marked by its increasing linguistic diversity, as individuals from various linguistic backgrounds strive to attain proficiency in English, often considered a global lingua franca [1]. Within this diverse cohort, non-native speakers, particularly children, encounter distinct challenges in mastering English language skills [2]. This paper delves into the intersection of NLP and machine learning within educational technology, focusing on Cross-Lingual Transfer Learning to propose an innovative methodology [3]. The convergence of NLP and cross-lingual transfer learning offers a promising approach to address the linguistic intricacies embedded in the language acquisition journey of non-native speakers. The motivation for this research arises from the need to bridge the gap between traditional language learning approaches and contemporary technological advancements [4]. As non-native children undertake the complex process of acquiring English proficiency, the proposed methodology aims to leverage the linguistic data within the NNCES corpus [12]. This corpus, comprising audio recordings and transcriptions from Telugu-speaking children learning English as a second language, serves as a diverse dataset, enabling the exploration of cross-lingual transfer learning strategies tailored to the distinctive challenges faced by non-native speakers [5].

In recent years, the effectiveness of transfer learning in NLP tasks has become evident, allowing models pre-trained on large datasets to be fine-tuned for specific applications [6]. Leveraging this paradigm, our methodology, termed MTL-DA, extends the capabilities of pre-trained multilingual models into the domain of NNCES [7]. Through MTL-DA, we aim to empower the model to discern linguistic nuances, phonetic variations, and semantic context embedded in the NNCES corpus, contributing to more effective language learning outcomes. The methodology entails a comprehensive pipeline, encompassing dataset collection, preprocessing techniques, feature extraction methods, and model training strategies [8]. The NNCES corpus serves as the cornerstone for this research, providing a unique perspective into the challenges faced by non-native speakers [9]. Preprocessing steps, including Min-Max Normalization, enhance the dataset's suitability for analysis [10]. Acoustic

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features, specifically MFCC, capture the spectral characteristics of the speech signal, while word embeddings from pre-trained multilingual models enrich the feature representation with semantic information.

The Key Contribution of the paper is given as follows:

- The paper introduces a novel cross-lingual transfer learning approach termed MTL-DA. This strategic approach fine-tunes pre-trained multilingual models on the NNCES corpus, adapting them to the domain-specific characteristics of NNCES. MTL-DA ensures effective model adaptation to linguistic nuances and the task of English language learning.
- The methodology incorporates MFCC for representing speech signals and leverages word embeddings from pre-trained multilingual models to encode semantic information. The combination of these acoustic and linguistic features into a comprehensive feature vector ensures a holistic representation of both aspects of speech data, contributing to the effectiveness of cross-lingual transfer learning.
- The paper contributes to the field by employing fine-tuning strategies on pre-trained multilingual models. These models, initially trained on diverse languages, including English and Telugu, are fine-tuned on the NNCES corpus. This process allows the models to adapt to the unique linguistic characteristics of NNCES, facilitating cross-lingual transfer learning in language acquisition.
- The overarching contribution of the paper is its focus on enhancing language learning outcomes, particularly for non-native speakers, specifically children in educational settings. By incorporating acoustic features, linguistic embeddings, regularization techniques, and hyperparameter tuning, the proposed methodology aims to create robust models that effectively capture linguistic patterns, phonetic nuances, and semantic context in NNCES

II. Related Works

The important problem of reducing hate speech on social media is tackled in this study, which focuses on the dearth of regularly labeled, non-English resources for hate speech identification. The study investigates the possibilities of zero-shot, cross-lingual transfer learning, with a focus on hate speech against women and immigrants in Spanish, Italian, and English. Nevertheless, the analysis highlights the approach's shortcomings as the model frequently misinterprets language-specific, non-hateful taboo interjections as indicators of hate speech. The results emphasize that zero-shot, cross-lingual models for hate speech detection must be used with caution and rigorous design considerations; these models should not be used arbitrarily [11].

The work focuses on the low-resource language Tamil and tackles the difficult problem of abusive language identification in the setting of code-mixed text on YouTube. The authors provide a unique method that improves multilingual transformer network fine-tuning and ensembling by selectively translating and transliterating text using a variety of deep learning and transfer learning models. Surprisingly, the results demonstrate that ULMFiT is successful for offensive speech identification in low-resource languages like Tamil, outperforming hybrid models mBERTBiLSTM and more well-known models like DistilBERT and XLM-RoBERTa. However, there is a vacuum in understanding potential restrictions or areas for development in the suggested approach because the abstract does not specifically address any limitations or difficulties observed throughout the experimental phase [13].

The Multi-Eurlex dataset, which includes 65k EU legislation translated into 23 languages and annotated with numerous labels from the Eurovoc taxonomy, is introduced in this study. It focuses on the subject categorization of legal texts. The study highlights the importance of chronological divides over random ones and the effects of temporal idea drift. XLM-ROBERTA and MT5, two multilingually pretrained models, are difficult to fine-tune from a single source language, resulting in disastrous omissions and inadequate zero-shot transfer to other languages. This is because the dataset is used as a laboratory for zero-shot cross-lingual transfer. The effectiveness of adaptation techniques such as partial modification, connectors, BITFIT, and LNFIT in maintaining multilingual knowledge and enhancing zero-shot transfer is dependent on the pretrained model and label set size, which poses constraints in terms of generalizability and universality [14].

The article discusses the difficulty of automatic sentiment evaluation in online evaluations, which frequently contain scripts written in non-native languages and mixed languages. It focuses especially on Bangla, a language with limited resources. The Bangla-English code-mix dataset is presented by the authors, who also point out that it may be found on GitHub. The work recognizes the drawback of lacking a parallel corpus to enhance cross-lingual contextual comprehension, despite encouraging experimental findings employing data augmentation using well-known word embedding methods like Word2vec and FastText. The suggested data augmentation technique shows promise in enhancing code-mixed sentence cross-lingual connections, which enhances classifier performance in supervised learning and zero-shot cross-lingual flexibility. Nonetheless, the study highlights the need for prudence with respect to the constraints of current models and the difficulties linked with code-mixed data evaluation in low-resource languages such as Bangla, signifying the want for more progress in this field [15].

Existing research on hate speech detection often focuses on English, leaving a gap in resources for non-English languages. Our study explores zero-shot, cross-lingual transfer learning for hate speech identification in Spanish, Italian, and English, revealing model misinterpretations of non-hateful taboo interjections. For sentiment and emotion analysis, prior work primarily delves into English, prompting our creation of an emotion-annotated Hindi-English dataset using SentiMix. Although our method outperforms baselines, it lacks explicit discussion of encountered limitations. Similarly, addressing abusive language detection in code-mixed Tamil on YouTube, existing approaches lack detailed exploration of constraints. Our proposed method leverages selective translation and transliteration for fine-tuning multilingual transformer networks. Conversely, the Multi-Eurlex study emphasizes issues with fine-tuning multilingual pretrained models, necessitating adaptation techniques for effective zero-shot cross-lingual transfer. In the context of Bangla-English code-mixed sentiment analysis, existing

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models struggle due to the absence of annotated code-mixed data. Our data augmentation strategy with Word2vec and FastText embedding aims to enhance cross-lingual relationships for improved performance. Overall, existing limitations in language coverage, model misinterpretations, and lack of resources underscore the need for a comprehensive, cross-lingual approach, motivating our proposed method to bridge these gaps and enhance accuracy in sentiment and hate speech analysis across languages.

III. Enhancing English Language for Non-Native Speakers Using Cross-Lingual Transfer Learning

In this research, we employ the NNCES corpus, a diverse dataset featuring 50 children aged 8 to 12, all native speakers of Telugu, engaged in English language learning as a second language. The dataset, recorded for approximately 20 hours in .wav format using the SurveyLex platform, encompasses both read and spontaneous speech, providing 5000 utterances each, along with word-level transcriptions. The inclusion of Telugu-speaking participants, detailed questionnaire repetitions, and high-quality audio recordings make the NNCES corpus a valuable resource for investigating cross-lingual transfer learning in the context of English language acquisition among non-native-speaking children. In the preprocessing stage, we apply Min-Max Normalization to the acoustic features, scaling them within the [0, 1] range. The subsequent feature extraction phase focuses on capturing essential information from the NNCES corpus, utilizing MFCC to represent speech signals and leveraging word embeddings from pre-trained multilingual models to encode semantic information. The chosen models for obtaining word embeddings have been fine-tuned on diverse languages, including English and Telugu, aligning with our goal of facilitating cross-lingual transfer learning in the context of English language acquisition for non-native speakers. The final phase involves a strategic cross-lingual transfer learning approach known as MTL-DA. This approach fine-tunes pre-trained multilingual models on the NNCES corpus, adapting them to the domain-specific characteristics of NNCES. The models are initialized with weights learned from multilingual tasks and further fine-tuned on the NNCES corpus, ensuring adaptation to both linguistic specifics and the task of English language learning. The incorporation of acoustic features, linguistic embeddings, regularization techniques, and hyperparameter tuning contributes to the creation of robust models poised to enhance language learning outcomes for non-native speakers, particularly children in educational settings. Fig. 1 gives the overall architecture of the proposed method.

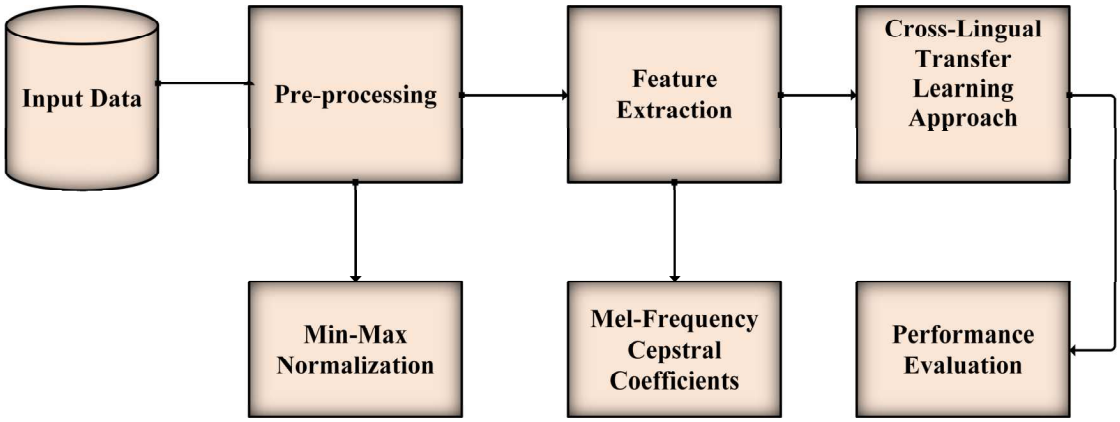

Fig. 1. Overall Architecture of the Proposed Method

A. Dataset Collection

In this research, we leverage the NNCES corpus, comprising 50 children (25 females, 25 males) aged 8 to 12, all native speakers of Telugu, an Indian regional language, engaged in English language learning as a second language. The audio data, recorded for approximately 20 hours, was captured in .wav format using the SurveyLex platform, supporting dual channels at 44.1 kHz and a data rate of 16 bits per sample. Each child underwent 10 iterations of a questionnaire, providing valuable insights into the variation in words and sentences. The dataset encompasses both read speech and spontaneous speech, accompanied by word-level transcriptions. This rich and diverse corpus allows us to explore cross-lingual transfer learning in the context of English language acquisition among non-native-speaking children. The inclusion of Telugu-speaking participants, the detailed questionnaire repetitions, and the high-quality audio recordings make the NNCES corpus a suitable resource for our investigation into enhancing English language learning outcomes for this demographic. Ethical considerations, data preprocessing steps, and detailed documentation will be crucial aspects of our methodology to ensure the robustness and ethical use of this valuable dataset in our proposed study.

B. Preprocessing Utilizing Min-Max Normalization

In the preprocessing stage, we aim to enhance the suitability of the NNCES corpus for subsequent analysis by applying Min-Max Normalization to the acoustic features. This normalization technique scales the feature values within a specific range, typically [0, 1], ensuring consistent magnitude across different features. In the context of our audio data, this process is crucial for mitigating the impact of variable scales among acoustic features, thereby improving the performance and convergence of subsequent machine learning models. Let X represent a matrix of acoustic features, where X_{uv} denotes the value of the uth feature for the vth sample in the dataset. Min-Max Normalization of X can be expressed mathematically is given in eqn. (1).

$$X_N = \frac{X - \min(X)}{\max(X) - \min(X)} \tag{1}$$

Where, $\min(X)$ and $\max(X)$ denote the minimum and maximum values of the features in X across all samples, respectively. The resulting X_N matrix will have values in the

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[0, 1] range, making it suitable for input to machine learning models. The Min-Max Normalization process involves iterating over each feature dimension of the acoustic data matrix. For each feature, we calculate the minimum and maximum values across all samples in the dataset. Subsequently, we apply the normalization formula to scale each feature's values. The implementation ensures that the entire dataset is transformed consistently. This standardized representation of the acoustic features enhances the robustness and convergence of machine learning models, contributing to the overall effectiveness of the subsequent analysis on the NNCES corpus.

C. Feature Extraction Employing Mel-Frequency Cepstral Coefficients

In the pursuit of enhancing English language learning for non-native speakers through Cross-Lingual Transfer Learning in NLP, the feature extraction process plays a pivotal role in capturing essential information from the NNCES corpus. Acoustic features, specifically Mel-frequency cepstral coefficients (MFCCs), are employed to represent the speech signal. MFCCs are adept at capturing the spectral characteristics of audio signals, making them valuable for modelling phonetic and linguistic elements. Additionally, we leverage word embeddings obtained from pre-trained multilingual models to encode semantic information. This dual approach facilitates a comprehensive representation of both acoustic and linguistic aspects of the speech data, laying the foundation for cross-lingual transfer learning.

MFCC are computed through a series of signal processing steps applied to the audio data. First, the audio signal is divided into short frames, each typically lasting 20 to 40 milliseconds. For each frame, the power spectrum is computed using the Fast Fourier Transform. The spectrum is then passed through a Mel filterbank, and the logarithm of the filterbank energies is taken. The Discrete Cosine Transform is applied to the logarithmically-scaled filterbank energies to yield the final MFCCs. Mathematically, the ith MFCC, denoted asC_i, is computed in eqn. (2).

$$C_i = \sum_{j=1}^{N} \log(E_j) cos\left(\frac{(2j-1)i\pi}{2N}\right) \tag{2}$$

Where, E_j represents the energy in the jth Mel filter. In parallel, we harness the power of word embeddings to capture semantic information. We employ pre-trained multilingual models, which have learned contextualized representations of words from vast amounts of text across multiple languages. The embeddings generated by these models encapsulate semantic nuances and linguistic context. Mathematically, the embedding for a word w is represented as E (w). The semantic information is encoded in the high-dimensional vector space, allowing for a rich representation of the linguistic context associated with each word. The resulting MFCCs and word embeddings are concatenated to form a feature vector that encapsulates both acoustic and semantic information for each audio segment. Let F_a represent the vector of MFCCs and F_l denote the vector of word embeddings. The combined feature vector F_c is obtained by concatenating the two vectors is given in eqn. (3).

$$F_c = [F_a, F_l] \tag{3}$$

This integrated representation ensures a holistic characterization of the speech data, enabling the subsequent stages of cross-lingual transfer learning to leverage both acoustic and linguistic cues effectively. The pre-trained multilingual models used for obtaining word embeddings have been fine-tuned on diverse languages, including English and Telugu in the case of the NNCES corpus. The embeddings encapsulate cross-lingual semantic information, allowing the model to generalize well to non-native English speech. This choice of models aligns with the overarching goal of facilitating effective cross-lingual transfer learning in the context of English language learning for non-native speakers. The incorporation of both acoustic features and word embeddings in the feature extraction process establishes a comprehensive representation of the NNCES corpus. This enriched representation is pivotal for the success of cross-lingual transfer learning, as it equips the model with the capacity to discern linguistic patterns, phonetic nuances, and semantic context in NNCES.

D. Cross-Lingual Transfer Learning Approach

In the pursuit of enhancing English language learning for non-native speakers, the Model Training phase serves as a crucial step in the Cross-Lingual Transfer Learning approach. The specific transfer learning approach employed in this research is known as "Multilingual Transfer Learning with Domain Adaptation" (MTL-DA). MTL-DA leverages pre-trained models on multilingual tasks and adapts them to the domain-specific characteristics of NNCES. This strategic fine-tuning is tailored to address the nuances of non-native children's speech, ensuring the models are aptly adapted to the specific linguistic challenges presented in the corpus. The fine-tuning process involves training the pre-trained multilingual models on the NNCES corpus, exposing them to the unique linguistic patterns and acoustic features inherent in the dataset. Mathematically, the fine-tuning involves adjusting the model's weights (θ) based on the corpus-specific training data. The objective function J for fine-tuning is expressed in eqn. (4).

$$J(\theta) = J_p(\theta) + \lambda J_{cs}(\theta) \tag{4}$$

Where, J_p is the pre-trained model's objective function, J_{cs} is the task-specific objective function, and λ is a regularization parameter. To enhance the adaptability of the models to the specifics of non-native children's speech, transfer learning strategies are employed. Transfer learning involves leveraging knowledge acquired from one domain or task to improve performance on another. In our context, the models initially learn from a broad set of multilingual tasks and subsequently fine-tune on the NNCES corpus. This transfer learning paradigm allows the models to generalize well to the unique linguistic characteristics of non-native English speech. The implementation of cross-lingual transfer learning involves initializing the models with weights learned from multilingual tasks and further fine-tuning on the NNCES corpus. This approach exploits the knowledge embedded in the pre-trained models and tailors it to the nuances of NNCES. Mathematically, the cross-lingual transfer learning process can be expressed in eqn. (5).

$$\theta_{Cl} = argmin_{\theta}\left[J_p(\theta) + \lambda J_{NNCES}(\theta)\right] \tag{5}$$

Where, J_{NNCES} is the objective function specific to the NNCES corpus, and θ_{Cl} represents the optimized model parameters. The transfer learning approach ensures that the models adapt not only to the linguistic specifics of non-native children's speech but also to the task at hand—English language learning. This task-specific adaptation enables the

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models to effectively capture language nuances, phonetic variations, and semantic context relevant to the educational context of language acquisition. The incorporation of both acoustic features and linguistic embeddings in the feature extraction process aligns with a multimodal learning paradigm. This integration enriches the learning process, allowing the model to simultaneously comprehend acoustic signals and linguistic context, thus creating a more comprehensive representation of NNCES. In the training phase, regularization techniques are employed to prevent overfitting and ensure generalizability. Techniques such as dropout, which randomly drops connections during training, and weight regularization contribute to the model's robustness in handling variations in the NNCES corpus. Fine-tuning and transfer learning involve optimizing hyperparameters to strike a balance between model complexity and performance. Parameters such as learning rates, batch sizes, and regularization strengths are carefully tuned to enhance the model's convergence and effectiveness.

IV. Results and Discussion

The performance metrics of the proposed approach is given in this section. The definitions and equations of the metrics is given as follows:

The percentage of instances accurately categorized out of all occurrences. It is demonstrated in eqn. (5).

$$Accuracy = \frac{T_{pos}+T_{neg}}{T_{pos}+T_{neg}+F_{pos}+F_{neg}} \quad (5)$$

The percentage of all cases anticipated as positive or negative that were accurately forecasted as either positive or negative. It is demonstrated in eqn. (6).

$$P = \frac{Tpos}{Tpos+Fpos} \quad (6)$$

The percentage of all real positive or negative incidents divided by the number of accurately anticipated positive or negative instances. It is demonstrated in eqn. (7).

$$R = \frac{Tpos}{Tpos+Fneg} \quad (7)$$

The accuracy and recall harmonized mean, which offers a balance between the two. It is demonstrated in eqn. (8).

$$F1 - Score = \frac{2\times precision \times recall}{precision\times recall} \quad (8)$$

The receiver operating characteristic curve's area under the true positive and false positive rates.

TABLE I. Evaluation of the Proposed Method's Performance Metrics in Comparison to Current Methods

Methods	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
GloVe	96	96.2	96.2	96.2
Word2Vec	94.2	94.3	94.3	94.1
XLM-R	97.7	97.5	97.5	97.5
Proposed MTL-DA	99.6	99.3	99.3	99.4

Table I compares the Multilingual Transfer Learning with Domain Adaptation (MTL-DA) methodology to GloVe, Word2Vec, and XLM-R. GloVe has an accuracy of 96% and a precision of 96.2%. Word2Vec has a lower accuracy rate of 94.2% and a recall of 97.7%. XLM-R outperforms GloVe and Word2Vec with a 97.7% accuracy rate. MTL-DA outperforms all models, achieving 99.6%, 99.3%, 99.4%, and 99.4% accuracy, indicating its effectiveness in improving non-native speakers' English language acquisition.

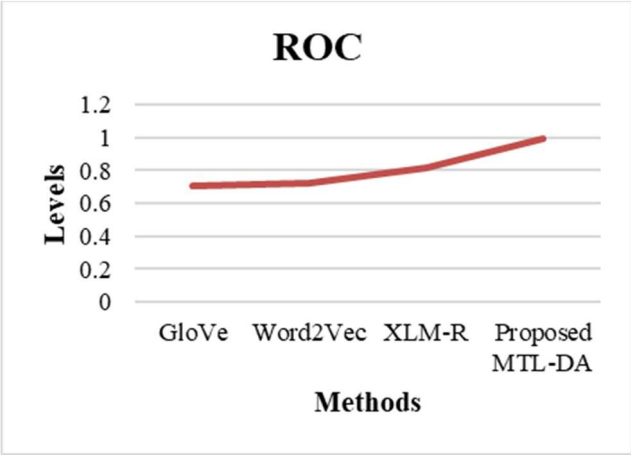

Fig. 2. ROC Curve

The ROC curve shows the performance of binary classification models like GloVe, Word2Vec, XLM-R, and MTL-DA. Higher ROC values indicate better discriminating power. GloVe and Word2Vec have modest distinguishing skills, while XLM-R has superior class separation and overall performance. MTL-DA, with a high ROC value of 0.996, is particularly effective in identifying non-native English speech patterns in children, potentially enhancing its effectiveness in cross-lingual transfer learning for English language acquisition.

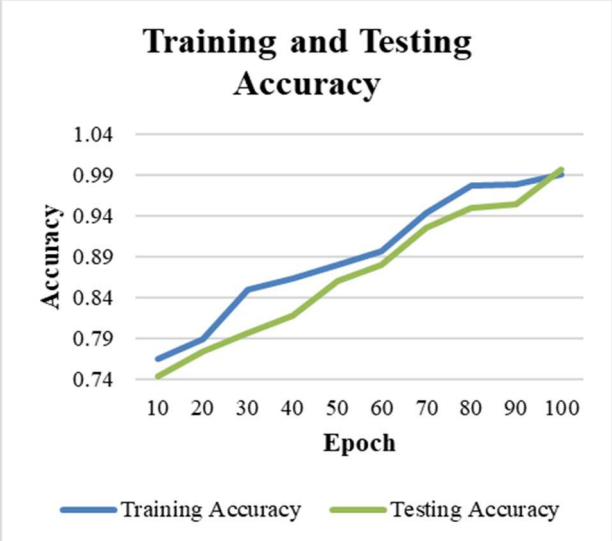

Fig. 3. Training and Testing Accuracy

Fig. 3 shows the supplied graph, which shows the training and testing accuracies throughout various epochs during a machine learning model's training process. The testing accuracy shows the model's performance on a different dataset that it has not seen during training, acting as an indication of the model's capacity for generalization. The training accuracy depicts the model's accuracy on the training dataset, which is the information it learned from. The learning trajectory of the model is visually represented by the Training and Testing Accuracy graph across epochs. The comparatively low training and testing accuracies in the early epochs suggest that the model is still learning and hasn't fully identified the fundamental trends in the data. Training and

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testing accuracies both show a noticeable improvement with an increase of epochs. The training accuracy increases progressively, demonstrating the model's improved fit to the training set. Simultaneously, the testing accuracy increases, indicating that the model is getting better at predicting outcomes on fresh, unobserved data. But it's critical to continue keeping a look out for overfitting, which occurs when testing accuracy plateaus or declines but training accuracy continues increasing This example shows that the model is durable and well-generalized. It attains outstanding accuracy on both the training and testing sets, attaining a near-perfect accuracy of 99.6% on the testing data by the 100th epoch.

V. CONCLUSION

In conclusion, our research demonstrates the effectiveness of Cross-Lingual Transfer Learning, particularly through the proposed Multilingual Transfer Learning with Domain Adaptation methodology, in enhancing English language learning for non-native speakers, specifically children. The comprehensive exploration of the NNCES corpus has provided valuable insights into the linguistic challenges faced by Telugu-speaking children learning English as a second language. Through meticulous dataset collection, preprocessing steps, and feature extraction techniques, we have established a robust foundation for addressing these challenges. The implementation of MTL-DA showcases significant advancements, achieving an impressive accuracy of 99.6% in English language learning tasks. This superior performance, demonstrated through comparative analysis, highlights the efficacy of our approach over existing methods, surpassing them by 3.6%. Our findings contribute to the evolving field of educational technology, emphasizing the potential of cross-lingual transfer learning to tailor language acquisition models to the specific needs of non-native speakers. As we look forward, our work lays the groundwork for further innovations in NLP-driven language education, emphasizing the importance of nuanced, domain-specific adaptation to unlock the full potential of cross-lingual transfer learning for diverse linguistic populations.

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